



LESSON - 7



SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

Data Manipulation Language (DML)

"DML handles the data in your database—adding, updating, and deleting records. It's how you manage and work with your data!"

Command	Description	Syntax
INSERT	Insert data into a table	INSERT INTO table_name (column1, column2, ...) VALUES (value1, value2, ...);
UPDATE	Update existing data within a table	UPDATE table_name SET column1 = value1, column2 = value2 WHERE condition;
DELETE	Delete records from a database table	DELETE FROM table_name WHERE condition;
LOCK	Table control concurrency	LOCK TABLE table_name IN lock_mode;
CALL	Call a PL/SQL or JAVA subprogram	CALL procedure_name(arguments);

INSERT - Add Data to Tables

Used to insert new records into a table.

```
INSERT INTO table_name (column1, column2, column3, ...)  
VALUES (value1, value2, value3, ...);
```

Example :

```
INSERT INTO Customers (CustomerName, ContactName, Address, City, PostalCode, Country)  
VALUES ('Cardinal', 'Tom B. Erichsen', 'Skagen 21', 'Stavanger', '4006', 'Norway');
```

```
INSERT INTO Customers (CustomerName, ContactName, Address, City, PostalCode, Country)  
VALUES  
( 'Cardinal', 'Tom B. Erichsen', 'Skagen 21', 'Stavanger', '4006', 'Norway'),  
( 'Greasy Burger', 'Per Olsen', 'Gateveien 15', 'Sandnes', '4306', 'Norway'),  
( 'Tasty Tee', 'Finn Egan', 'Streetroad 19B', 'Liverpool', 'L1 0AA', 'UK');
```

UPDATE - Modify Data to Tables

The UPDATE statement is used to modify the existing records in a table.

```
UPDATE table_name  
SET column1 = value1, column2 = value2, ...  
WHERE condition;
```

Example :

```
UPDATE Customers  
SET ContactName = 'Alfred Schmidt', City= 'Frankfurt'  
WHERE CustomerID = 1;
```

WARNING : Be careful when updating records. without **WHERE** ALL records will be updated!

DELETE - Delete Data to Tables

The DELETE statement is used to delete existing records in a table.

```
DELETE FROM table_name WHERE condition;
```

Example :

```
DELETE FROM Customers  
WHERE CustomerId=1;
```

```
DELETE FROM Customers;
```

WARNING : Be careful when deleting records. without WHERE ALL records will be Deleted!

